

Ninth day of class - 9/23

Fun fact: Andrew mentioned that $9272025 = 3045^2$ and $27092025 = 5205^2$ are both squares. (This is Saturday's date read in both the mm/dd/yyyy and dd/mm/yyyy formats).

I think this sort of thing (where $mm \neq dd$) has only happened once before: 1030225 and 3010225 are both squares.

The next time it will happen is on April 22, 3025, as

$$4223025 = 2055^2 \quad 22043025 = 4695^2.$$

Last time: Magma demo.

Today: An introduction/review of group theory.

What is a group?

A group G is a set that also has a binary operation $*$ on it. (This is a “way of combining” two elements of G .) We need to have

- i) For all $a, b \in G$, that $a * b \in G$.
- ii) For all $a, b, c \in G$, that $(a * b) * c = a * (b * c)$. (The associative law.)
- iii) There is an $e \in G$ so that $a * e = e * a = a$ for all $a \in G$. (e is called the identity. We will sometimes write it as 1_G , sometimes as 0 if we write the group additively.)
- iv) For all $a \in G$, there is an $a^{-1} \in G$ (called the inverse of a) so that $a * a^{-1} = e$.

Note: For a general group, it need not be the case that $a * b = b * a$ for all $a, b \in G$. If a group has this property, it is called *abelian* (after the Norwegian mathematician Niels Abel). All the groups we work with in this course will be abelian, but I'll give you some examples of non-abelian groups.

Examples:

- i) $\mathbb{Z}/n\mathbb{Z} = \{0, 1, \dots, n-1\}$. The operation $*$ is defined by $a * b = (a + b) \bmod n$. The identity is 0 and the inverse of a is $n - a$.
- ii) $\mathbb{Z}$, the set of all integers. We define $a * b = a + b$. The identity is 0 , and the inverse of m is $-m$.
- iii) $\text{GL}_2(\mathbb{R})$, the set of all 2×2 matrices $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ so that $\det(M) = ad - bc \neq 0$. We define $A * B = AB$, the product of the two matrices.

To prove that $A * B \in \text{GL}_2(\mathbb{R})$, we need to know that $\det(AB) \neq 0$. This follows because $\det(AB) = \det(A) \det(B)$.

The identity is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, and the inverse of M is

$$\frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Note that $\text{GL}_2(\mathbb{R})$ is not abelian, since $MN \neq NM$ in general.

iv) $\mathbb{Q}^\times$, the group of nonzero rational number. The operation $a * b = ab$ is multiplication. The identity is 1, and the inverse of a is $1/a$. (We can also define $\mathbb{R}^\times$ to be the nonzero real numbers.)

v) $(\mathbb{Z}/n\mathbb{Z})^\times = \{a : 0 \leq a \leq n-1 \text{ with } \gcd(a, n) = 1\}$. The group operation is $a * b = ab \bmod n$. The identity is 1, and the inverse of a is the multiplicative inverse of $a \bmod n$.

vi) $E(\mathbb{Q})$. If E is an elliptic curve, then $E(\mathbb{Q})$ is the set of rational points on E with the operation that $P + Q$ is obtained by drawing the line through P and Q , finding the third intersection point, and reflecting.

Questions?

Subgroups:

If G is a group and $H \subseteq G$, we say that H is a subgroup if the group operation G makes H into a group. In order for this to happen we need to check that

- (i) H is non-empty,
- (ii) if $a, b \in H$, then $a * b \in H$,
- (iii) if $a \in H$, then $a^{-1} \in H$.

(Q: Why is that enough?)

Note: In general, $(a * b)^{-1} = b^{-1}a^{-1}$. This rule may be familiar when you work with matrices.

Example 1: If $G = \mathbb{Z}$, let $H = 2\mathbb{Z}$ be the set of all even numbers. Then H is non-empty, because $0 \in H$.

Also, if $a, b \in H$, then a and b are both even, and so $a + b$ is also even and so $a + b \in H$.

If $a \in H$, then $a^{-1} = -a$ is also even and so it is in H .

Example 2: Let $G = \mathbb{Q}^\times$ and $H = (\mathbb{Q}^\times)^2 = \{a^2 : a \in G\}$. This set is nonempty (since $1 \in H$), and if $h_1, h_2 \in H$, then $h_1 = a^2$ and $h_2 = b^2$ and so $h_1 * h_2 = a^2b^2 = (ab)^2 \in H$. Also, if $h = a^2$ with $a \in \mathbb{Q}^\times$, then $h^{-1} = 1/a^2 = (1/a)^2 \in H$.

Finite generation:

If $g_1, \dots, g_n$ are elements of G , then we write $\langle g_1, g_2, \dots, g_n \rangle$ for the smallest subgroup H of G that contains all the elements $g_1, g_2, \dots, g_n$. If $H = G$, we say that G is generated by $g_1, \dots, g_n$.

If G can be generated by a finite number of elements, we say that G is finitely generated.

if G can be generated by a single element, we say that G is cyclic.

Orders: If G is a group, $|G|$ is the number of elements in G . If $g \in G$ is an element, we say that the order of g is the *smallest* positive integer k so that $g^k = 1_G$ is the identity of G . If the order of g is 1, then $g = 1_G$. If there is no such k , we say that the order of g is infinite.

[If G is a group, we can let $H = \{g^k : k \in \mathbb{Z}\}$. This is a subgroup of G , and the order of H is equal to the order of g .]

Example: Let $E : y^2 = x^3 - 13392x - 1080432$. It is not so obvious, but this elliptic curve has the point $P = (168, 1188)$ on it. We find that

$$2P = (564, -13068)$$

$$3P = (564, 13068)$$

$$4P = (168, -1188)$$

$$5P = (0 : 1 : 0).$$

Thus, the point P has order 5 in $E(\mathbb{Q})$. In fact, we didn't need to compute all the way to $5P$ to see that. Once we compute $3P$, we see that $3P = -2P$. This makes $5P = \mathcal{O}$.

For this E , it turns out that $E(\mathbb{Q})$ consists only of these five points, that is $E(\mathbb{Q}) = \{P, 2P, 3P, 4P, \mathcal{O}\}$.

Fact about orders: If the g has order n , then g^m has order $\frac{n}{\gcd(m,n)}$. So every power of g has order dividing n .

Homomorphisms:

A homomorphism is a function $\phi : G \rightarrow H$ between two groups that “plays nice with the group operations on G and H ”. The definition is that ϕ is any function that must satisfy

$$\phi(a * b) = \phi(a) * \phi(b).$$

(Note that $a, b \in G$ and so $a * b \in G$. However, $\phi(a)$ and $\phi(b)$ are in H and so $\phi(a) * \phi(b) \in H$.)

Example 1: If $G = \text{GL}_2(\mathbb{R})$ and $H = \mathbb{R}^\times$, we can define $\phi : G \rightarrow H$ by $\phi(M) = \det(M)$. This is a homomorphism because

$$\phi(a * b) = \phi(ab) = \det(ab) = \det(a) \det(b) = \phi(a) * \phi(b).$$

Q: Why? What good is a homomorphism? Why are they important?

A: Many good homomorphisms are functions that gives you something simpler and more concrete to think about than what you had before. Most of you would agree that a 2×2 matrix is harder to think about than just a nonzero real number.

Later in the course, we will construct a homomorphism from $E(\mathbb{Q})$ to a simpler group (which we haven't defined yet) that will allow us to study in more detail the structure of the group $E(\mathbb{Q})$.

Isomorphisms:

An isomorphism between two groups G and H is a homomorphism $\phi : G \rightarrow H$ that is one-to-one and onto. An isomorphism pairs up elements of G with elements of H , and the homomorphism property

$$\phi(a * b) = \phi(a) * \phi(b)$$

means that the elements are paired up with each other in a way that respects the group law. (That is if $a, b \in G$ are paired with c and d in H , then $a * b$ is paired up with $c * d$.)

This means that the two groups G and H are group-theoretically “the same”. We write $G \cong H$.

For example, if $E : y^2 = x^3 - 13392x - 1080432$, then the map $\phi : \mathbb{Z}/5\mathbb{Z} \rightarrow E(\mathbb{Q})$ given by $\phi(k) = kP$ is an isomorphism, where $P = (168, 1188)$.

Direct products:

If G and H are groups, $G \times H$ is the group whose set is $\{(g, h) : g \in G, h \in H\}$, the collection of all ordered pairs of elements in G and elements in H . This set can be made a group by defining

$$(g_1, h_1) * (g_2, h_2) = (g_1 * g_2, h_1 * h_2).$$

The identity is $(1_G, 1_H)$, and the inverse of (g, h) is (g^{-1}, h^{-1}) .

Why is this construction useful?

Theorem. *If G is a finitely-generated abelian group, then is isomorphic to a direct product of (some finite number) of cyclic groups.*

Theorem (Mordell). *If $E/\mathbb{Q}$ is an elliptic curve, then $E(\mathbb{Q})$, the group of all rational points on E , is finitely generated.*

Example: If $E : y^2 = x^3 - 2x$, then $E(\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}$. If $\phi : \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z} \rightarrow E(\mathbb{Q})$ is the isomorphism, then $\phi((1, 0)) = (0, 0)$, and $\phi((0, 1)) = (2, 2)$.

This means that the point $P = (0, 0)$ has order 2, and the point $Q = (2, 2)$ has infinite order. Also, every point in $E(\mathbb{Q})$ can be written uniquely in the form $aP + bQ$ where $a \in \{0, 1\}$ and $b \in \mathbb{Z}$.

For example, the point $R = (-1/169, 239/2197)$ is on E . Since ϕ is onto, there is an element $(a, b) \in \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}$ with $\phi((a, b)) = R$. In fact, $\phi((1, 3)) = R$. This means that $\phi((1, 3)) = \phi((1, 0) + (0, 3)) = \phi((1, 0)) + \phi((0, 3)) = \phi((1, 0)) + 3\phi((0, 1)) = P + 3Q$.

More theorems about elliptic curves, stated in the language of groups

We can restate Mordell’s theorem in the following form.

Theorem. *If $E/\mathbb{Q}$ is an elliptic curve, then there is a non-negative integer r and a finite abelian group T so that*

$$E(\mathbb{Q}) \cong \overbrace{\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}}^{r \text{ copies}} \times T.$$

The number r is called the *rank* of $E(\mathbb{Q})$. If $r = 0$, then $E(\mathbb{Q})$ is finite, and if $r \geq 1$, then $E(\mathbb{Q})$ is infinite. We often write

$$\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z} = \mathbb{Z}^r.$$

Example: If $E : y^2 = x^3 - x^2 - 24649x + 1355209$, then $E(\mathbb{Q})$ has rank 4. (This curve as found by Wiman in 1945.) We have

$$E(\mathbb{Q}) \cong \mathbb{Z}^4 \times (\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}).$$

Generators are $P_1 = (-675/4, 6601/8)$, $P_2 = (4375/9, -275356/27)$, $P_3 = (-175, -528)$, $P_4 = (1511/25, -35916/125)$, $T_1 = (67, 0)$, $T_2 = (113, 0)$.

The Magma command `MordellWeilGroup(E)` returns an abelian group G and an isomorphism between G and $E(\mathbb{Q})$. (Or rather, it tries to return this info.)

Warning: We don't know if there is an algorithm that will compute the rank r of $E(\mathbb{Q})$.

Theorem (Mazur, 1977). *The only possibilities for T (up to isomorphism) are*

- $\mathbb{Z}/1\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z}, \mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/5\mathbb{Z}, \mathbb{Z}/6\mathbb{Z}, \mathbb{Z}/7\mathbb{Z}, \mathbb{Z}/8\mathbb{Z}, \mathbb{Z}/9\mathbb{Z}, \mathbb{Z}/10\mathbb{Z}, \mathbb{Z}/12\mathbb{Z}$, and
- $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$, and $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$.

Theorem (Nagell-Lutz). *If $E : y^2 = x^3 + ax^2 + bx + c$ and $P = (x_0, y_0) \in E(\mathbb{Q})$ has finite order, then $x_0 \in \mathbb{Z}$ and $y_0 \in \mathbb{Z}$.*

If $E/\mathbb{Q}$ is an elliptic curve, we can reduce the curve mod p . We can take the geometric means of doing the group law on E , translate that into algebra, and do that algebra mod p .

The net effect is that for each prime p , we get a homomorphism $\phi_p : E(\mathbb{Q}) \rightarrow E(\mathbb{F}_p)$. (Well, we only get such a homomorphism if E is actually an elliptic curve over $\mathbb{F}_p$.) The Nagell-Lutz theorem above guarantees that if P is a point of finite order, then $\phi_p(P) \neq (0 : 1 : 0) \in E(\mathbb{F}_p)$ for any prime p .

Consequence: If $E(\mathbb{Q})$ contains a point of order k , then $E(\mathbb{F}_p)$ contains a point of order k for all primes p (where $E/\mathbb{F}_p$ is non-singular).

Example: Let $E : y^2 = x^3 - 16x + 16$. What is T ?

Claim: $T = \{(0 : 1 : 0)\}$.

Suppose not. Suppose that $P \in E(\mathbb{Q})$ has finite order and $P \neq (0 : 1 : 0)$. Then $\phi_3(P) \in E(\mathbb{F}_3) \cong \mathbb{Z}/7\mathbb{Z}$ but it cannot have order 1. Thus, $\phi_3(P)$ has order 7. Then $7\phi_3(P) = \phi_3(7P) = (0 : 1 : 0)$. We know that $7P$ has to have finite order (because P does), but the Nagell-Lutz theorem shows that the only way $7P$ can reduce to the point at infinity mod 3 is if it equals $(0 : 1 : 0)$. Thus, $7P = (0 : 1 : 0)$ and so P has order 7.

On the other hand, $E(\mathbb{F}_5) \cong \mathbb{Z}/8\mathbb{Z}$ and so $\phi_5(P)$ must have order that divides 7. But the only element of $E(\mathbb{F}_5)$ with order dividing 7 is $(0 : 1 : 0)$ and this implies that $\phi_5(P) = (0 : 1 : 0)$. But this contradicts the Nagell-Lutz theorem.

So there aren't any points of finite order other than $(0 : 1 : 0)$.

(The command `TorsionSubgroup(E)` in Magma will also return the finite abelian group T and an isomorphism between T and the *subgroup* of points of finite order in $E(\mathbb{Q})$.)